Reference Event 3: The Louisiana Superdome as Refuge of Last Resort in Hurricane Katrina, August and September 2005

Characterization Against the Commercial Facilities Sector Design Basis

DBA-CF-RE-3 — Reference Event Instrument, DBA-CF Series

Lineage: Issued under the Commercial Facilities Sector Framework v0.5; DBA-MA Parent Framework; conferred on the proposal of Public Assembly and Mass-Gathering Venues (DBA-CF-PA-FC1)

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose, Scope and Position

1.1 Purpose

This instrument reads the use of the Louisiana Superdome as the refuge of last resort in New Orleans during and after Hurricane Katrina, in August and September 2005, back through DBA-CF-PA-FC1 (*Public Assembly and Mass-Gathering Venues*), whose category of loss of the refuge it characterizes (§5.5), at the class's third reference facility, the stadium as refuge (§1.3). It asks whether the class's design basis anticipated what happened, and returns what it finds.

1.2 Position and Scope

A reference event instrument, in the series convention, reads a real event backward through the design basis, adds no requirements of its own, and returns findings and recommendations for disposition. This one is issued under the Commercial Facilities Sector Framework v0.5, inherits the DBA-MA parent framework, and was conferred by the framework's §9.1 on the author's decision of 14 October 2026, on the proposal of *Public Assembly and Mass-Gathering Venues*. Section references of the bare form §N are to that class at v0.1, the version this instrument read.

The scope runs from the opening of the Superdome as a refuge before landfall on 29 August 2005 to the departure of its last evacuees on Saturday 3 September, and takes in the Convention Center only as the refuge that the Superdome's overflow became. The coordination of the response, the absence of command at both sites and the mass-care mission are the Emergency Services series' and are read in its instrument. The House and Senate committees' findings are stated as their own. Officials whose decisions they examined are named by role.

The venue in its disaster-support role lies within this class. The Foundational Definitions are cited by designation, FD-REG holding their versions: FD-DBL (*Design Basis Layers*) bears on Section 5.

2. Sources and Attribution

2.1 Sources and Their Limit

The primary sources are the House select bipartisan committee's report *A Failure of Initiative* of February 2006, its chapter on shelter and housing in particular, and the Senate Committee on Homeland Security and Governmental Affairs' report *Hurricane Katrina: A Nation Still Unprepared* of May 2006, its executive summary in particular. The departure of the last evacuees and the National Guard's count on the Friday are from contemporary reporting.

2.2 Where the Sources Differ

The figures for the number sheltered differ. The Senate committee gives a population of more than twenty thousand; the House committee's account, as *The Failure of Coordination in*

Hurricane Katrina (DBA-ESS-RE-8) reads it, more than ten thousand; and an officer of the Guard was reported on the Friday as putting it at about twenty thousand. Each is its source's, and the counts were taken at different times. Neither committee's report as read gives a figure for deaths in the Superdome, and this instrument states none. The Senate committee's finding that tens of thousands waited until as late as the Saturday to be evacuated is stated for New Orleans as a whole, not for the Superdome alone.

3. The Event

3.1 The Choice of Refuge

On the House committee's findings, Louisiana's plans required a shelter of last resort to be outside the floodplain, or able to place its people on floors above the flood, and to be a hurricane-wind-resistant structure. The Superdome was used as one although it lay in the floodplain. The committee found that state and local governments made inappropriate selections of shelters of last resort.

3.2 The Storm and the Building

On the Senate committee's findings, the storm tore the covering from the Superdome's roof, caused leaking and cut the power, leaving the plumbing, air conditioning and public address system useless. The House committee found that the roof's extensive wind damage showed that the building was not a hurricane-wind-resistant structure. More than twenty thousand people stayed in the Superdome without working plumbing for nearly a week; portable toilets were requested for it and never arrived.

3.3 Overflow

By the Tuesday afternoon, on the Senate committee's findings, the Superdome was overcrowded and officials turned further people away. A population of about nineteen thousand gathered at the Convention Center, which, the House committee found, was never planned as a shelter and became one out of necessity and improvisation; supplies did not reach it until the Friday, and the Guard evacuated it on the Saturday within eight hours.

3.4 Evacuation

On the Senate committee's findings, the head of the federal emergency management agency assured the state on the Monday that five hundred buses were on their way and would arrive within hours. The agency did not direct the Department of Transportation to send buses until very early on the Wednesday; they did not begin to arrive until the Wednesday evening, nor in significant numbers until the Thursday. In the meantime transit buses carried people not out of the city but to the Superdome. Its last evacuees left on Saturday 3 September, as reported.

4. Technical Decomposition

The event decomposes into initiating elements (I-1 to I-3) and effect elements (E-1 to E-4), as Table 4-1 sets out. Composed, they are the class's loss of the refuge (§5.5) at the stadium as refuge, in Chains R and L.

Table 4-1. Event elements and their mapping. Section references are to *Public Assembly and Mass-Gathering Venues* v0.1.

No.	What happened	Class provision	Section
I-1	Refuge of last resort in the floodplain, not wind-resistant	Not designated for a hazard it cannot withstand (§4.6)	3.1
I-2	Roof covering torn; power, plumbing and address system lost	Shelter sustainment (§3.3, §4.6)	3.2
I-3	Buses promised Monday, directed Wednesday, arriving Wednesday to Thursday	Time and means to evacuate (§4.6, §5.5)	3.4
E-1	More than twenty thousand held nearly a week without working plumbing	Chain R; Chain L	3.2
E-2	Overcrowded by Tuesday; people turned away	Shelter capacity (§3.3)	3.3
E-3	About nineteen thousand at an unplanned refuge, supplied on Friday	Chain R; overflow	3.3
E-4	Last evacuees out on Saturday 3 September	Second residual (§6.8)	3.4

5. Tier Determination

5.1 The Four Characteristics

There was no adversary. Intentionality, persistence, precision and denial are not present.

5.2 Determination

The event does not meet the threshold for the military-action extension, under the author's rulings of 10 and 13 October 2026 recorded at FD-DBL (*Design Basis Layers*) §6.4. It is recorded as a non-adversarial event realizing the class's §5.5 in Chains R and L. That is the tier determination returned for the DBA Reference Event Register. Emergency Services is proposed as an impacted sector, the mass-care mission and the evacuation being its, and Transportation for the buses on which the evacuation depended.

6. Characterization Against the Design Basis

6.1 §2.4 and §5.5 The Class's Account of the Event

Held, with a gap. The account holds against the committees' reports in the floodplain and wind-resistance findings, the roof, the power, the plumbing, the toilets and the buses. Erratum E-1: the

class states as a finding on the Superdome that tens of thousands waited in unspeakably horrible conditions until as late as the Saturday to be evacuated; the Senate committee states that finding for New Orleans as a whole, and the class should attribute it so, recording the Superdome's last departures on Saturday 3 September from contemporary reporting (Section 2.2). Two refinements are offered and are not errata: the figures for the number sheltered differ by source and time and may be stated as each source's; and the assurance of five hundred buses on the Monday states the means of evacuation the class reads.

6.2 §4.6 Not Designated for a Hazard It Cannot Withstand

Held. The class requires that a venue not be designated as a refuge for a hazard it cannot withstand, and that a refuge of last resort's shortfall be stated so that the mass-care plan does not count on it. The state's own plan stated the requirement; the Superdome did not meet it and was used anyway. The class was written from this event.

6.3 §3.3 and §4.6 Sustainment Independent of the Disaster

Held. The class requires a refuge to sustain its population with power, water, sanitation, food and medical care independent of the systems the disaster takes. The storm took the power and with it the plumbing, the air conditioning and the means of speaking to the people inside; the toilets requested from outside never came. The requirement reads the event.

6.4 §3.3 The Shelter Capacity and Its Overflow

Held, with a gap. The class states shelter capacity as the population a refuge can hold. By the Tuesday the Superdome was full and people were turned away, and about nineteen thousand gathered at a building never planned as a shelter, without supplies until the Friday. The class states the capacity; it does not require a refuge to state what happens when the capacity is reached, where those turned away go, and what that place can provide. Finding F-1.

6.5 §4.6 and §5.5 The Time and Means to Evacuate

Held, with a gap. The class requires the time and means to evacuate a refuge to be stated and not to depend on the transport the same disaster is likely to take. The buses that emptied the Superdome were not taken by the disaster; they came from outside it, and their dispatch waited on an order that was not given until the Wednesday, two days after an assurance that they were on their way. The class's requirement is met in form by means the disaster does not take; it is not met by means whose dispatch is not committed before the refuge opens. Finding F-2.

6.6 §6.8 The Second Declared Residual

Held. A refuge overwhelmed by a disaster beyond its design leaves a residual in the population sheltered, measured by that population. More than twenty thousand people were held for nearly a week in conditions the building could not sustain.

6.7 §7.3 The AI-Governance Design Basis

Not exercised. No component within the scope of *Layer Decomposition and Determinism* §4.3 bore on the event.

6.8 Summary of Design Basis Performance

Of the seven readings, three Held, with a gap, the class's account among them for its one erratum; three Held; and one was Not exercised. Nothing was Missed. The class's §4.6 and §5.5 were written from this event and read it. What the event adds is that a refuge's design basis includes where its overflow goes, and that its means of evacuation count only if their dispatch is committed before the refuge opens.

7. Findings

Section references are to *Public Assembly and Mass-Gathering Venues* v0.1 unless another instrument is named.

7.1 F-1 — A Refuge States Where Its Overflow Goes

The Superdome was overcrowded by the Tuesday and people were turned away; about nineteen thousand gathered at a building never planned as a shelter, without supplies until the Friday (Section 3.3). The finding is that the class should require a venue carrying the disaster-support overlay to state, with its shelter capacity, the point at which it stops admitting people, how that is decided and by whom, where those it cannot admit are directed, and what that place can provide and for how long; and should treat a refuge whose overflow has no stated destination as a shortfall to be stated for the mass-care plan under §4.6.

Affected: §2.2, §3.3, §4.6, §5.5. Disposition: R-1; noted to *Emergency Management and Coordination* (DBA-ESS-EMCo) for the mass-care plan.

7.2 F-2 — The Means of Evacuation Are Committed Before the Refuge Opens

Five hundred buses were assured on the Monday, ordered early on the Wednesday and arrived from the Wednesday evening, while transit buses carried people to the refuge (Section 3.4). The finding confirms §4.6 and is that the class should credit the means of evacuating a refuge only where they are committed before the refuge opens, with the trigger and the authority for their dispatch stated so that dispatch does not wait on a request made after the disaster strikes, and should state the time from that trigger to the refuge's emptying as part of its shelter sustainment.

Affected: §3.3, §4.6, §5.5, §6.9. Disposition: R-2; noted to *Emergency Management and Coordination* and to the Transportation series for the buses.

8. Recommendation Register

Priority is on the series' one-to-three scale, 1 the most consequential. R-1 and R-2 dispose to *Public Assembly and Mass-Gathering Venues*.

No.	Title	Affected	Recommendation	Pri.
R-1	State where the overflow goes	§2.2, §3.3, §4.6, §5.5	With shelter capacity, state the admission stop, who decides it, where those turned away go and what that place provides; a refuge without one states the shortfall. Noted to ESS-EMCo.	1
R-2	Commit the means of evacuation	§3.3, §4.6, §5.5, §6.9	Credit evacuation means only if committed before opening, with dispatch trigger and authority stated; state trigger-to-empty time. Noted to ESS-EMCo and TRAN.	1

9. Open Items

OI-1: the register row. Project 0 is asked to enter the tier determination of Section 5.2. OI-2: erratum E-1 is owed to the class at its next revision. OI-3: no figure for deaths in the Superdome is relied on; if a source the committees used is read, Section 2.2 records it as that source's.

10. Standing of This Document

10.1 What This Instrument Establishes

It modifies neither class nor framework. It states the committees' findings as their own and makes no finding of fault.

10.2 Relationship to the Class's Use Case

The class's use case traces §6.3, the refuge struck during a conflict, with the power, water and transport that sustain and would evacuate it denied. This event is that scenario's peacetime form: a refuge in the hazard's path, its power and sanitation lost, its overflow sent to an unplanned building, and its people held for nearly a week while the means to move them waited on an order. F-1 and F-2 give the use case its two quantities, the refuge's capacity with its overflow and the committed time to empty it. What an adversary would add is the use case's construction.

10.3 Relationship to Other Instruments

The Failure of Coordination in Hurricane Katrina (DBA-ESS-RE-8) reads the same days for the response, including the absence of command at the Superdome and the Convention Center. *The Loss of Power at Memorial Medical Center* (DBA-HPH-RE-2) reads the same storm at a hospital whose power failed with its population inside.

References

United States House of Representatives, Select Bipartisan Committee to Investigate the Preparation for and Response to Hurricane Katrina, *A Failure of Initiative*, February 2006, chapter on shelter and housing. United States Senate, Committee on Homeland Security and Governmental Affairs, *Hurricane Katrina: A Nation Still Unprepared*, May 2006, executive summary. Contemporary reporting of the Superdome's evacuation, NBC News, 2 and 3 September 2005. The DBA instruments named in the text, with the Foundational Definitions and FD-REG, and the DBA-MA Parent Framework.

Revision History

Version 0.1, 14 October 2026: issued in Project 11 under the Commercial Facilities Sector Framework v0.5, on the conferral of the framework's §9.1 of 14 October 2026, read against *Public Assembly and Mass-Gathering Venues* v0.1. Written new under the Reference Event Compression Standard v0.3. Tier determination at Section 5.2, under FD-DBL §6.4 as ruled on 10, 13 and 14 October 2026; F-1 and F-2, R-1 and R-2; erratum E-1 to the class.